

The Arrow's Deep Field: Dark-Matter Phenomenology from the Khronon, and the Unification of Event-Density Gravity

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Abstract

GR-II identified Event Density's gravitational class as **khronometric**: Einstein's tensor sector plus one scalar — the khronon, the substrate's arrow of time made dynamical, anchored to the cosmological rest frame. This paper shows that the same scalar naturally carries the **dark-matter phenomenology** of ED's published gravity arc, unifying the corpus's two gravity programs into one theory. The construction has one missing span and two built ends. The ends: the khronon (GR-II), and the MOND sector — $a_0 = cH_0/(2\pi)$ from the cosmic decoupling surface (Paper_029), the Combination Rule $a = \sqrt{a_N a_0}$ (Paper_030), BTFR (Paper_031), the MOND field equation (Paper_036). The span: the khronon's kinetic structure. Promoting the khronometric acceleration term to a function $W(A^2)$ of the congruence acceleration — the established khronometric-MOND structure — the static weak field reduces to the **modified Poisson equation in the metric potential itself**, $\text{grad} * [\mu_{\text{tot}}] \text{grad} \Phi = 4\pi G \rho$ with $\mu_{\text{tot}} = 1 + c_1 W'$. The high-acceleration limit is GR-I, untouched (commitment-linearity $\implies \mu \rightarrow 1$); the deep-infrared limit is **uniquely fixed by matching Paper_030** ($\mu_{\text{tot}} \rightarrow x$, forced-given-030), embedding the Combination Rule and BTFR relativistically — at the surfaced cost of an **infrared cancellation** (the khronon takes over the static constraint below a_0), standard in this class. Statics isolate the construction: $\theta = \sigma = 0$ for the aligned khronon, so the MOND carrier is the *unique* active foliation term, and the ED identity $a_i = 1/2 d_i \ln b$ reads the whole sector in substrate language — *the deep field switches on where bandwidth gradients fall below the cosmic rate*. **The historical kill-checks pass structurally**: lensing tracks the MOND potential with **no added vector field** (the modification is metric-borne; the conformal-coupling disease TeVeS patched never arises; slip is $O(\Phi)$ -suppressed); no new ghost (the W -sector adds no time derivatives in unitary gauge); the classic AQUAL ellipticity conditions hold; screening is PPN-safe in the linear khronometric corner. The one soft spot — the $A \rightarrow 0$ degeneracy — is resolved without retrofit: the degenerate point is **exact Minkowski vacuum, which is not in ED's state space** (the substrate always ticks), the pure- A^2 form is a truncation whose generic completion $W_{\text{cal}}(A^2, \theta)$ is healthy at the physical cosmological point, and an **orthogonality theorem** ($\theta = 0$ in statics) sequesters the regulator family from everything tested. The khronon's cosmological face ties the arc to SCBU — **one scale, H_0 , in four roles** (background rate, a_0 , horizon locus, R_H boundary) — and its infrared vacuum term is naturally of order $\rho_{\text{Lambda}} \sim H_0^2 c^2/G$, the observed dark-energy order, read as single-scale consistency (form-tier only). The result: **one khronometric gravity** — Einstein at high accelerations, MOND in the deep

field, dark energy’s order carried — in which “dark matter” is *the deep-infrared regime of the arrow’s own field*. No particle, no added vector, no new primitive. Clusters and the CMB remain owed.

1. Introduction

1.1 What this paper does

ED’s gravity corpus contained two programs. The published arc (Papers 025–038) derived Newton, the transition acceleration $\mathbf{a}_0 = c\mathbf{H}_0/(2\pi)$, the Combination Rule, and BTFR — the dark-matter-free account of galactic dynamics — covariantized in Paper_033 on a *postulated* metric with a *postulated* scalar. The post-pivot arc (GR-I/II) *derived* the metric (the bandwidth metric, with the weak-field Einstein tests) and *derived* the scalar (the khronon: the arrow made dynamical). This paper supplies the one missing identification: **the postulated scalar of the MOND arc is the derived scalar of the curvature arc**. It proceeds in four steps: the mapping (§3), the static weak-field reduction with the GR and MOND limits (§4), the lensing and viability checks (§5–§6), and the cosmological regulator with the SCBU tie-in (§7).

Three orienting statements, up front:

- **The hinge.** The paper’s only derivational hinge is the deep-infrared matching to Paper_030’s geometric-mean law, which **uniquely fixes** the non-analytic term in W' . No other freedom is exercised anywhere in the construction; everything else is inherited, derived, or labeled a family.
- **The economy.** No new primitive, field, or symmetry is introduced. The khronon was *counted* by GR-II’s mode analysis before MOND was on the table; this paper spends a field the theory already had.
- **What survives (preview).** The three historical kill-checks of relativistic MOND all pass structurally: **lensing** tracks the MOND potential with no added vector (§5); **no new ghost** arises (§6.1); **solar-system screening** is PPN-safe (§6.3). The one soft spot — the deep-vacuum degeneracy — is resolved at the level of ED’s state space and EFT generality (§7). And the cosmological regulator is a *family, sequestered from every static test*: its role is to make the theory well-posed on FRW backgrounds, not to influence galactic phenomenology (§7.3).

1.2 Why this matters

Three reasons. First, **unification**: the corpus’s two gravity lines fuse into one theory with one extra field — and that field was not added for the purpose; it was *counted* by GR-II’s mode analysis before MOND was on the table. Second, **the dark-matter claim sharpens**: “no dark matter; modified dynamics below $\mathbf{a}_0$ ” becomes “the galactic phenomenology is the deep-infrared regime of the preferred-foliation scalar” — a specific field with an independent origin (the arrow), independent constraints (GW170817, PPN, pulsars), and independent predictions (a breathing GW mode; wide-binary deviations). Third, **the historical failure modes are addressed structurally, not by patches**: the constructions that killed earlier relativistic MOND (conformal couplings blind to lensing; added vectors; fine-tuned $\mathbf{a}_0$) are each avoided by something ED already had.

Why the unification is nontrivial. Four scales that have no business coinciding — the rate of the arrow’s cosmological flow, the acceleration at which galactic dynamics turn non-Newtonian, the energy density of the vacuum, and the locus where horizons form — are here carried by **one scalar field with one background scale**. In any framework where these are independent sectors, their numerical coincidences ($a_0 \sim cH_0/2\pi$; $\rho_{\Lambda} \sim H_0^2 c^2/G$) are unexplained accidents. In this construction they are the *same number wearing four hats*, because the khronon has only one number to offer.

1.3 How this fits the arc

- **Papers 027–031, 036:** Newton; a_0 ; Combination Rule; BTFR; the MOND field equation. (*the deep-field end — inherited*)
- **Paper_033:** scalar-tensor covariantization on a postulated metric and scalar. (*fused here: the scalar is identified*)
- **Paper GR-I:** the bandwidth metric; weak-field Einstein tests. (*the high-acceleration end — inherited*)
- **Paper GR-II:** the khronometric class; the khronon; $c_T = c$; Lorentz safety. (*the field itself — inherited*)
- **Paper KM-I (this paper):** the unification — the khronon’s deep field is the MOND sector.

This completes the second of the two reconciliation items GR-I §7 declared open (the khronon $\leftrightarrow$ MOND relation); the first (the bandwidth $\leftrightarrow$ strain/flow substrate-route identification) remains open.

1.4 Conventions and regime of validity

Signature $(-, +, +, +)$; $c = 1$ where convenient. The derivations are **static, weak-field, quasi-spherical**, in the **continuum (DCGT) limit**; the cosmological statements (§7) are FRW-background structural results. **No strong-field, no time-dependent-lensing, no cluster, and no CMB claims are made.** Exact numerical coefficients in the reductions are convention-dependent and carried schematically (c_1 -type constants of order unity); the *structures* and *limits* are the results.

2. Primitive Inputs

Substrate primitives (postulated, Paper_087): P11/P13 (the arrow and the tick — the khronon’s origin, via GR-II), P04 (bandwidth — the metric and lapse, via GR-I), P02/P05 (adjacency/transport). The Σ -selection’s orientation-blindness and P-Commitment-Linear (GR-I) enter through the inherited results.

Inherited results (load-bearing):

- **GR-I:** $g_{ij} \sim b^{-1}$, $N^2 \sim b$, the weak-field Schwarzschild metric; P-Commitment-Linear $\Rightarrow$ the Einstein branch at high accelerations.
- **GR-II:** the khronometric class; the khronon T with $u_\mu = d_\mu T / \sqrt{X}$ hypersurface-orthogonal; the preferred frame = the cosmological rest frame; $c_T = c$; universal Lorentz violation.

- **Paper_029:** $a_0 = cH_0/(2\pi)$ (value anchored to H_0).
- **Paper_030:** the Combination Rule $a = \sqrt{a_N a_0}$ (the substrate’s own deep-field law, via the P14 bilocal coupling).
- **Papers_031/036:** BTFR; the MOND field equation (the flat-space form this paper’s static limit must and does land on).

External dictionary (inherited mathematics/physics, labeled I):

- **Khronometric/aether-MOND** (Blanchet–Marsat 2011; Zlosnik–Ferreira–Starkman 2007): a preferred-foliation scalar with non-canonical *acceleration* kinetics yields the MOND modified-Poisson equation in the static weak field, with the interpolation given by the kinetic function’s derivative. The class’s reported stability corners are cited as reported.
- **Classical AQUAL well-posedness** (Bekenstein–Milgrom): ellipticity conditions $\mu > 0, d(x\mu)/dx > 0$.

No new primitive is introduced; no primitive forcing is invoked. The paper’s single structural commitment beyond inheritance is the *typing* of the kinetic sector (the acceleration invariant, §3.2) — which is argued, not assumed, since the alternative typing fails independently.

2.5 Load-Bearing Step Audit

Status legend: **I** = inherited (from a cited paper or established external result); **D** = derived here; **D-structural** = derived from structure alone, no model details; **conditional** = derived given a labeled input; **family** = an honestly-unfixed functional freedom; **form-tier** = an order-of-magnitude/structural consistency, no value claimed; **open** = declared open.

Step	Status	Source / justification
Khronon exists; class is khronometric; cosmic-frame anchored	I	GR-II
Metric/lapse from bandwidth; Einstein branch at high A	I	GR-I
$a_0 = cH_0/2\pi$; Combination Rule; BTFR; MOND field eq	I	Papers 029/030/031/036
Khronometric-MOND dictionary ($W(A^2) \Rightarrow$ modified Poisson)	I	Blanchet–Marsat; ZFS
Kinetic invariant = the congruence acceleration (not $(dT)^2$)	D	§3.2 — the $(dT)^2$ typing is structurally inert or lensing-dead
Statics isolate the W-sector ($\theta = \sigma = 0$; khronon aligns)	D	§4.1 — kinematics of static congruences

Step	Status	Source / justification
ED identity $a_i = 1/2d_i \ln b$	D (identity)	§4.1 — via GR-I's $N^2 \sim b$
Modified Poisson $\text{grad} * [\mu_{\text{tot}} \text{grad } \Phi] = 4\pi G \rho, \mu_{\text{tot}} = 1 + c_1 W'$	D	§4.2 — form-level; constants conventional
GR limit ($W' \rightarrow \alpha$ small $\Rightarrow \mu \rightarrow 1$; GR-I untouched)	D	§4.3
Deep-IR limit $\mu_{\text{tot}} \rightarrow x \Rightarrow$ Combination Rule + BTFR	D conditional on I-030	§4.4 — forced-given-030 ; includes the IR cancellation
The IR cancellation ($W' \rightarrow -1/c_1 + x/c_1$)	surfaced cost	§4.4 — standard-in-class; stress-tested §6
Lensing tracks the MOND potential; no vector needed	D	§5 — metric-borne modification; slip $0(\Phi)$ -suppressed
No new ghost (W-sector adds no time derivatives)	D-structural	§6.1 — unitary-gauge counting
AQUAL ellipticity across the interpolation	D	§6.2 — classic conditions; deep IR passes
Screening PPN-safe; interpolation family	D / I	§6.3 — linear-khronometric corner; ephemeris bounds
Cassini-constrained τ -mode = the breathing scalar; benign as screened	D	§6.4
$(A, \theta) = (0, 0)$ outside ED's state space (no Minkowski vacuum)	D-structural	§7.1 — P11/P13; the khronon's background is the Hubble flow
Degeneracy = truncation artifact; $W_{\text{cal}}(A^2, \theta)$ generic completion	D (EFT-generality)	§7.2 — regulator family , labeled
Orthogonality: tested sector invariant under the regulator	D	§7.3 — $\theta == 0$ in statics
One scale in four roles; SCBU carrier	structural identification	§7.4
$\rho_{\Lambda} \sim H_0^2 c^2 / G$ vacuum-term resonance	form-tier only	§7.4 — no value derived; V1-reconciliation open
Čerenkov / BBN / FRW stability	filters on the family	§7.5
Primitives-level origin of the deep-IR branch	OPEN — guarded	preamble 2
Clusters / CMB	NOT ADDRESSED	preamble 8

All steps I, D, structural, conditional, family, form-tier, open, or not-addressed. *The single derivational hinge is forced-given-030; the two honest families (interpolation; regulator) are sequestered and independently constrained.*

3. The Mapping: One Missing Span, Two Built Ends

3.1 The ends

The field (GR-II). ED’s arrow lives in the law, forcing a preferred foliation; the propagating content is 2 tensor modes + 1 scalar — the khronon, whose background is cosmic time and whose background rate is the Hubble rate.

The law and the scale (Papers 028–031, 036). $a_0 = cH_0/(2\pi)$ from the decoupling surface; the Combination Rule $a = \sqrt{a_N a_0}$; BTFR; and the flat-space MOND field equation this construction’s static limit must reproduce.

3.2 The span, and its typing

The missing piece is the khronon’s kinetic structure. Its **typing is argued, not chosen**: a kinetic function of $(dT)^2$ fails twice over — in statics $(dT)^2 \sim 1 - 2\Phi - |\text{grad } \tau|^2$ depends on the *potential*, not its gradient (yielding screening-type physics, never $\mu(|\text{grad } \Phi|/a_0)$), and its conformally-coupled variant is the historical RAQUAL construction that fails lensing. The correct carrier is the **acceleration of the khronon congruence**, $a_\mu = \text{perp}_\mu \hat{n} \nabla_\nu \ln \sqrt{X}$ — and this stays inside the foliation EFT GR-II already counted: the general quadratic sector is $\lambda \theta^2 + \beta \sigma_{\mu\nu} \sigma^{\mu\nu} + a_0^2 W(A^2)$, $A^2 = \frac{a_\mu a^\mu}{a_0^2}$, and the MOND extension promotes **only the acceleration term**:

$$S_T = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R + \lambda \theta^2 + \beta \sigma_{\mu\nu} \sigma^{\mu\nu} + a_0^2 W(A^2) \right], \quad A^2 = \frac{a_\mu a^\mu}{a_0^2},$$

with matter coupled to $g_{\mu\nu}$ **only** (universal coupling — in ED, matter *is* substrate content; there is no conformal matter frame to mis-couple). W is carried as a general function; its two asymptotics are fixed in §4, the region between them is an honest family.

The scale’s structural carrier. The khronon’s vacuum is the Hubble flow, so the only rate available to normalize its local gradients is the cosmic rate: the comparison “local acceleration against cH_0 ” is not a model choice but the field’s only option. **a_0 is the khronon’s background scale** — the Milgrom coincidence $a_0 \sim cH_0/2\pi$ acquires a carrier (the coefficient $1/2\pi$ stays inherited from Paper_029).

4. The Static Weak-Field Reduction

With the typing fixed (§3.2), the reduction proceeds in three steps: **alignment** (the khronon locks to static time), **isolation** (statics switch off every foliation term except the MOND carrier), and **reduction** (the modified Poisson equation, with its two limits).

4.1 Alignment, isolation, and the Bandwidth–Acceleration Lemma

Static, time-reversal-symmetric sources align the khronon ($T = t$; the misalignment τ is odd under the symmetry), making its congruence the static observers', whose kinematics are textbook: $\theta = 0$, $\sigma = 0$, $a_i = d_i \ln N$. Two consequences:

1. **Statics isolate the MOND carrier.** The λ and β sectors are inert ($\theta = \sigma = 0$); the acceleration function is the *unique* active foliation term in statics. The MOND modification and the static modification are the same term — by kinematics, not tuning. (The β -sector governs the tensor speed: $c_T = c$ from GR-II is untouched by W .)
2. **The Bandwidth–Acceleration Lemma.** GR-I derived $N^2 \sim b$, so

$$a_i = \partial_i \ln N = \frac{1}{2} \partial_i \ln b :$$

the khronon's acceleration is the logarithmic bandwidth gradient. This identity — referenced as such below — is the substrate reading of the entire MOND sector: the deep field switches on *where bandwidth gradients fall below the cosmic rate*.

4.2 The modified Poisson equation

With $g_{00} = -(1+2\Phi)$ and $A^2 = |\text{grad } \Phi|^2/a_0^2$ at leading order, varying the action gives (schematic order-unity constants flagged as conventional):

$$\nabla \cdot \left[\mu_{\text{tot}} \left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho, \quad \mu_{\text{tot}}(x) = 1 + c_1 W'(x^2)$$

— the AQUAL form, **in the metric potential itself** (not an auxiliary scalar). Here and throughout, $x = |\text{grad } \Phi|/a_0$. The Ψ -equation receives the khronon's stress, whose anisotropic part is second order in the potentials (§5).

4.3 The GR limit

At high accelerations $W' \rightarrow \alpha$, a small constant — the standard quadratic khronometric coupling, PPN-bounded. Then $\mu_{\text{tot}} \rightarrow 1$, $\Phi = \Psi$ at leading order, and **GR-I's weak-field Schwarzschild sector — the factor-of-two bending, redshift, precession — is untouched.** The residual $O(\alpha)$ preferred-frame effects are GR-II §8's standing α_1 , α_2 front, unchanged. ED-side, this limit is enforced: P-Commitment-Linear demands the Einstein branch at strong gradients — the screening regime is GR-I, not an option.

4.4 The deep-IR limit — forced-given-030, with the cost stated

Matching the corpus's derived deep-field law ($\mu_{\text{tot}} \rightarrow x$ so that, by Gauss-law integration, $|\text{grad } \Phi| = \sqrt{a_N a_0}$) fixes the deep-IR kinetic behavior **uniquely**:

$$W'(x^2) \xrightarrow{x \rightarrow 0} -\frac{1}{c_1} + \frac{x}{c_1} + \dots$$

Two pieces with two statuses. The non-analytic linear piece is the MOND carrier — the $X^{\{3/2\}}$ -type term, forced-given-030. The constant $-1/c_1$ deserves its own box:

The IR Cancellation Requirement. In the convention where Einstein–Hilbert keeps its standard normalization, the khronon’s kinetic slope must approach the constant $-1/c_1$ in the deep infrared — **exactly cancelling the Einstein gradient term in the static constraint below a_0 and taking it over.** Three facts about it: (i) it is **standard in this class** (the negative-branch kinetic function of generalized-aether MOND); (ii) it is the construction’s **only delicate point** — everything else is inheritance, kinematics, or sequestered family; (iii) it is **stress-tested in §6**, where it is shown *not* to destabilize statics (the total operator stays healthy even where $W' < 0$) and *not* to introduce a ghost.

With the requirement in place, the Combination Rule and BTFR slope-4 ($v^4 = GMa_0$) are **embedded relativistically** — the khronon carries Paper_030’s law; it does not re-derive it.

5. The Lensing Verdict

The check. Galactic lensing must track the MOND potential — the historical killer of relativistic MOND.

Why this construction is on the right side of history. The modification is **metric-borne**: §4.2 MONDifies the *metric potential*; matter and light couple universally to $g_{\{\mu\nu\}}$; there is no conformal scalar sector. RAQUAL failed lensing because its scalar coupled conformally and conformal factors do not deflect light — the *same* conformal-blindness fact GR-I §6 used to exclude Nordström. TeVeS’s vector field was a patch for that disease; **the khronon never has the disease the patch was for.**

The slip. The khronon’s anisotropic stress is $\sim W' a_{\{\lvert\angle i\}a_{\{j\}\rangle} = 0(|\text{grad } \Phi|^2)$ — second order in the potentials; even deep in the MOND regime its ratio to the leading terms is $0(\Phi) \sim 10^{-6}$ for galaxies. Hence $\Phi = \Psi \backslash, [1 + 0(\Phi)]$, both potentials are the MONDified potential, and the deflection is $\sim 2 \int \text{grad } _\perp \text{perp } \Phi_{\{\rm MOND\}}$:

Galactic lensing tracks the MOND potential, with no added vector field.

The kill-check passes at leading order — for the same structural reason ED’s gravity produced the factor of two: space and time potentials move together.

Caveats: leading-order, static; cluster-scale lensing inherits the MOND shortfall (preamble 8); the slip suppression presumes the healthy perturbations §6 addresses.

6. Viability: Ghosts, Ellipticity, Screening, and the τ -Mode

6.1 No new ghost (structural)

In unitary gauge $a_i = d_i \backslash \ln N$ — spatial gradients of the lapse only — so the W -sector contributes **no time derivatives at any order.** The scalar’s time-kinetic structure is the λ , β sector, untouched by the MOND generalization: **the no-ghost condition of the unified theory is the linear khronometric one.** (The W -sector can feed the effective

kinetic matrix through constraint elimination; that feedback is exactly the §7 degeneracy item, not a second pathology.)

6.2 Static well-posedness (the classic conditions)

The perturbed static operator has the AQUAL coefficients; the Bekenstein–Milgrom conditions $\mu_{\text{tot}} > 0$ and $d(x, \mu_{\text{tot}})/dx > 0$ hold across the interpolation — deep IR ($\mu = x$) passes both. **The infrared cancellation does not destabilize statics: $W' < 0$ there, but the *total* operator is healthy.**

6.3 Screening and the constrained family

Screening is the smallness of the residual, not Vainshtein: at solar accelerations ($x \sim 10^{-7}$ – 10^{-8}) the force correction is the saturation tail plus $O(\alpha)$ preferred-frame terms. Two channels, kept distinct: the **alpha_1, alpha_2 channel** (the linear corner; pulsar-tested; the standing GR-II falsification front, unchanged) and the **interpolation-tail channel**, where Cassini-class ephemeris bounds **exclude slow-saturating families** (inverse-power tails at the edge of exclusion; exponential tails survive). The interpolation is thereby a *constrained* family — measured, not protected.

6.4 The τ -mode

In unitary gauge the khronon perturbation is eaten: it **is** the breathing scalar GR-II already counted — not a new player. Benign as screened at high accelerations (dipole radiation suppressed with the scalar charge); its deep-IR dynamics (bars, mergers, satellites, wide binaries) are the theory’s *predictive* sector. Its only pathology channel is the §7 degeneracy — one soft spot, one address.

7. The Regulator and the Cosmological Face

7.1 The degenerate point is not in ED’s state space

The pure- A^2 sector degenerates as $A \rightarrow 0$ (vanishing stiffness, slowing modes) — but the configuration $(A, \theta) = (0, 0)$ at which the structure fails is **exact Minkowski vacuum, eternal and inert — and ED’s substrate forbids exact Minkowski vacuum: the tick never halts**. The khronon’s background is the Hubble flow; commitment does not cease (P13, P11). The degenerate point sits on the excluded boundary of the substrate’s state space. This is a domain observation, not a modification of W — and it is one of the construction’s strongest structural facts: *the substrate’s refusal to ever stop ticking is what protects its gravity theory*.

7.2 The truncation diagnosis and the regulator family

$W(A^2)$ was a truncation: the foliation EFT’s invariants are **a, theta, sigma**, and nothing singles out the acceleration as the sole argument of the IR structure. The general sector $\mathcal{W}(A^2, \Theta)$ ($\Theta = \theta/3H_0$) is **non-degenerate at the physical near-vacuum** ($A \rightarrow 0$, $\Theta \sim 1$), where the expansion supplies the stiffness. The Θ -form is **not forced** — the **regulator family**, labeled.

7.3 The orthogonality theorem

$\theta = 0$ for static congruences, so **every Theta-term vanishes identically in statics: the entire tested sector of this paper (§4–§6) is invariant under the regulator choice.** The family is sequestered — it cannot be tuned to help the galactic physics (no leak-back), and the galactic physics cannot vouch for it (no protection). Two independently falsifiable sectors.

7.4 The SCBU tie-in, and one tiered resonance

The khronon has exactly one background scale, and it appears in four roles: the background rate (H_0), the transition acceleration ($a_0 = cH_0/2\pi$), the horizon locus ($b \rightarrow 0$, GR-I), and SCBU’s unified boundary ($R_H = c/H_0$). **One scale, four roles: the SCBU boundary acquires a dynamical carrier — its field is the khronon.** And the IR sector’s vacuum term is naturally of order

$$\rho_\Lambda \sim \frac{a_0^2 c^2}{G} \sim \frac{H_0^2 c^2}{G}$$

— the observed dark-energy order: the long-noticed $\Lambda \sim a_0^2$ coincidence read as **single-scale consistency** (the khronon’s only scale is the cosmic rate, so its vacuum term can only be that size). Tiered hard: form/order-of-magnitude only; the $O(1)$ constant is not derived; $\Lambda \sim H_0^2$ is a consistency relation, not a prediction; and the reconciliation with the corpus’s V1-backreaction Λ (Paper_038.5) is open.

7.5 Filters on the family

The regulator family carries its own constraints: **BBN** (the $\lambda\theta^2$ sector renormalizes cosmological G ; bounded), **gravitational Čerenkov** (admissible members must give $c_s \sim O(1)$ at the cosmological point — the bare truncation fails this in voids, independently confirming §7.2’s diagnosis), and **FRW stability**. Pinning a member is model-building, outside this paper — the same epistemic boundary GR-II drew for the khronon parameters.

8. Position Relative to the Published Arc — the Fusion

- **Paper_033’s scalar is the khronon.** The pre-pivot arc *constructed* a scalar Φ on a postulated metric (both labeled postulates in its own audit); the post-pivot arc *derived* a scalar with exactly the required role and origin. The two gravity programs fuse: Paper_036’s MOND field equation is this paper’s §4.2 static limit; Paper_033’s covariantized equation is its covariant form, with the postulated background replaced by GR-I’s metric.
- **Paper_033/034’s “deep-MOND superluminality cost” is the khronon’s known class character** — what makes universal horizons exist — and not a causal pathology given the foliation: a flagged cost becomes an expected feature.
- **GR-I §7’s open item #2 (khronon $\leftrightarrow$ MOND) is hereby closed;** open item #1 (bandwidth $\leftrightarrow$ strain/flow) remains.

- **“Dark matter” in ED:** the galactic phenomenology is carried by the same field that carries the arrow of time — no particle, no added vector, no new primitive; the field was counted (GR-II) before it was needed (here).
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9. The Wedge — Empirical Content

1. **Wide binaries near $\mathbf{a}_0$** — the live observational frontier where this theory and Newtonian expectations genuinely diverge; an active measurement program, and the cleanest near-term discriminator.
 2. **Dynamical MOND** — the τ -mode’s excitation in time-dependent galactic systems is where MOND-class dynamics differ from particle dark matter; predictive territory, not yet computed.
 3. **The breathing GW polarization** (GR-II’s scalar mode) — now identified as the *same* field that does the galactic work: one detection channel, two phenomenologies.
 4. **The constrained families as shrinking falsifiable space** — Cassini on the interpolation; BBN/Čerenkov/FRW on the regulator: the theory’s unfixed sectors are being measured, not protected.
 5. **What is *not* claimed as a wedge:** clusters and the CMB (owed); the value of Λ (form-tier only); solar-system deviations beyond the standing α_1 , α_2 front.
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10. Falsification Criteria

10.1 Lensing slip in the MOND regime

Falsifier sentence: *Observation of a leading-order gravitational slip ($\Phi \neq \Psi$ beyond $O(\Phi)$ corrections) in galactic lensing relative to dynamics would falsify §5’s metric-borne mechanism — the construction has no vector field to repair it with.*

10.2 The Combination-Rule embedding

Falsifier sentence: *Galactic data inconsistent with $\mathbf{a} = \sqrt{\mathbf{a}_N \mathbf{a}_0}$ in the deep field (or BTFR slope-4 with the inherited $\mathbf{a}_0$) falsifies the deep-IR matching — and with it the unification, since the deep-IR form is forced-given-030.*

10.3 Wide binaries

Falsifier sentence: *Wide-binary samples near $\mathbf{a}_0$ showing purely Newtonian behavior at the precision where the deep-IR branch predicts deviation would falsify the unscreened sector.*

10.4 The interpolation family closes

Falsifier sentence: *Ephemeris (Cassini-class) bounds tightening to exclude all interpolation members compatible with the two fixed asymptotics would falsify the construction’s middle — there is no mechanism here to evade the static sector’s own equation.*

10.5 The AQUAL conditions

Falsifier sentence: *A demonstration that $\mu_{\rm tot} > 0$ or $d(x\mu_{\rm tot})/dx > 0$ must fail somewhere on the physical interpolation would falsify static well-posedness (§6.2).*

10.6 The regulator family empties

Falsifier sentence: *A proof that no $W_{\rm cal}(A \sim 2, \Theta)$ member simultaneously satisfies BBN, Čerenkov, and FRW-stability filters while remaining Θ -trivial in statics would falsify the §7 resolution of the degeneracy — returning the $A \rightarrow 0$ soft spot to the theory’s ledger as unresolved.*

10.7 Preferred-frame parameters (standing)

Falsifier sentence: *The GR-II falsification front (α_1, α_2 above bounds once the rule is pinned) carries over unchanged: the unification adds no protection against it.*

11. Position Statement

Event-Density gravity is one khronometric theory, and its dark-matter phenomenology is the deep-infrared regime of the khronon — the arrow of time made dynamical. Given the inherited ends — the khronon and its cosmic anchoring (GR-II), the metric and lapse (GR-I), the scale $a_0 = cH_0/2\pi$ (Paper_029) and the deep-field law (Paper_030) — the khronometric acceleration sector with a general kinetic function reduces, in the static weak field, to the AQUAL equation in the metric potential; the high-acceleration limit is GR-I (enforced by commitment-linearity); the deep-infrared limit embeds the Combination Rule and BTFR (forced-given-030, with the infrared cancellation surfaced); lensing tracks the MOND potential with no added vector; no new ghost arises; the classic ellipticity conditions hold; screening is PPN-safe with a Cassini-constrained interpolation family; the $A \rightarrow 0$ degeneracy is resolved at the level of ED’s state space (no Minkowski vacuum) and EFT generality (the sequestered regulator family); and the khronon’s cosmological face carries SCBU’s one scale in four roles, with a vacuum term naturally of the observed dark-energy order (form-tier).

What this paper claims. The unification: the corpus’s MOND sector and its curvature sector are one khronometric theory; the postulated scalar of Paper_033 is the derived khronon of GR-II; the deep-field law is embedded, the historical kill-checks (lensing foremost) pass structurally, and the open sectors are honest, sequestered, and independently constrained families.

What this paper does not claim. The MOND exponent is not derived from primitives (forced-given-030); a_0 and G are inherited; no value of Λ is derived; no interpolation or regulator member is selected; the class’s stability corners are cited, not re-verified; clusters and the CMB are not addressed; and the primitives-level origin of the deep-infrared branch — *why* the substrate’s response goes geometric-mean below the cosmic rate — remains the deepest declared open question.

Series context. Paper KM-I of the post-pivot curvature-emergence line, companion to GR-I and GR-II — and the fusion point with the published gravity arc (Papers 025–038). The remaining work is phenomenology (pin the rule; pin the families; compute the dynamical MOND sector) and the guarded origin question.

KM-I completes the unification GR-I and GR-II opened: **one scalar, one scale, one gravity.**

Appendix: Cross-References, Glossary, Reader Map

Cross-references

- **Paper_087:** position paper (13 primitives).
- **Papers_028/029/030/031/036:** decoupling surface; a_0 ; Combination Rule; BTFR; MOND field equation. (*the deep-field end*)
- **Paper_033/034:** scalar-tensor covariantization; deep-MOND superluminality. (*fused/reframed here*)
- **Paper GR-I:** bandwidth metric; lapse; weak-field Einstein tests. (*the high-acceleration end*)
- **Paper GR-II:** the khronometric class; the khronon; GW and Lorentz safety. (*the field*)
- **SCBU:** the $R_H = c/H_0$ boundary — here acquiring its dynamical carrier.

Glossary

- **Khronon.** The preferred-foliation scalar (GR-II): ED's arrow made dynamical; background = cosmic time.
- **Congruence acceleration a_μ .** $\text{perpgrad}_\mu \ln \sqrt{X}$; the kinetic invariant of the MOND sector.
- **Bandwidth–Acceleration Lemma.** In statics, $a_i = d_i \ln N = 1/2 d_i \ln b$ — the khronon's acceleration *is* the logarithmic bandwidth gradient (via GR-I's $N^2 \sim b$). The substrate reading of the MOND sector.
- **$W(A^2) / W_{cal}(A^2, \Theta)$.** The acceleration kinetic function and its regulated completion; $\mu_{\text{tot}} = 1 + c_1 W$.
- **Forced-given-030.** The deep-IR form is uniquely fixed by matching the corpus's independently-derived Combination Rule — an embedding, not a primitives derivation.
- **The infrared cancellation.** $W' \rightarrow -1/c_1$ as $A \rightarrow 0$: the khronon takes over the static constraint below a_0 ; standard-in-class, the construction's delicate piece.
- **Orthogonality theorem.** $\Theta = 0$ in statics $\Rightarrow$ the regulator family is invisible to everything tested.
- **Regulator family.** The Θ -sector of W_{cal} : open, sequestered, filtered by BBN/Čerenkov/FRW stability.
- **One scale, four roles.** H_0 as the khronon background rate, a_0 , the $b \rightarrow 0$ horizon, and the SCBU boundary.
- **Deep field.** The khronon's $A \ll 1$ regime — ED's account of dark-matter phenomenology.

Reader map and open work

Where to look next. - *The metric and the classical tests:* GR-I. — *The class, GW speed, Lorentz safety:* GR-II. - *The deep-field laws this paper embeds:* Papers 029/030/031/036. - *The cosmological face and the boundary:* SCBU.

Open work (declared): the primitives-level origin of the deep-IR branch (guarded); pinning

the interpolation and regulator families (model-building; Cassini/BBN/Čerenkov/FRW filters active); the dynamical MOND (τ -mode) sector; the V1-backreaction $\leftrightarrow$ khronon-vacuum reconciliation for Λ ; the bandwidth $\leftrightarrow$ strain/flow substrate-route identification (GR-I §7 item #1); clusters and the CMB (owed; address named — the **Theta**-sector's cosmology).

End of Paper KM-I.

Post-pivot curvature-emergence line, fusion point with the published gravity arc. The khronon — the arrow made dynamical — carries ED's dark-matter phenomenology: GR at high accelerations (enforced), the Combination Rule and BTFR below $\mathbf{a}_0$ (embedded, forced-given-030), lensing correct with no added vector (metric-borne), no new ghost, PPN-safe, with the $\mathbf{A} \rightarrow 0$ degeneracy resolved at the state-space and EFT-generality level and the regulator sequestered by the orthogonality theorem. One scale (H_0) in four roles ties the theory to SCBU, and the khronon's vacuum term is naturally of the dark-energy order (form-tier). No particle, no vector, no new primitive. Clusters/CMB owed; origin question guarded.